Roll No. : .....

MID-SEMESTER EXAMINATION MARCH-2025

Name of the Program: M. Tech **Semester: 2nd**

Name of the Course: Advanced Software Engineering **Course Code: MCS-201**

Time : 90 Minutes **Maximum Marks : 50**

Note:

i. Answer all the questions by choosing any one of the sub-questions
ii. Each question carries 10 marks

Q1	(10 Marks)	
(a)	Explain the key activities in software engineering and how they contribute to the software development lifecycle.	CO1
	OR	
(b)	What are the key architectural design patterns used in modern software systems?	CO3
Q2	(10 Marks)	
(a)	Explain the concept of socio-technical systems in software engineering. How do they differ from purely technical systems, and why is understanding socio-technical aspects important in software development?	CO5
	OR	
(b)	Describe the role of Object-Oriented Design (OOD) in modern software development. How do concepts like encapsulation, inheritance, and polymorphism improve software modularity and reusability?	CO4
Q3	(10 Marks)	
(a)	Discuss the challenges faced in software engineering and the essential skills required for a software engineer to overcome these challenges.	CO1
	OR	
(b)	Differentiate between functional and non-functional requirements in software engineering. Provide examples of each and explain their impact on software design.	CO3
Q4	(10 Marks)	
(a)	Explain how different software process models (Waterfall, Agile, Spiral) address the software development lifecycle. When should each model be used?	CO2
	OR	
(b)	Fault tolerance is a critical feature of distributed systems. Explain the key fault tolerance mechanisms used in distributed architectures.	CO4
Q5	(10 Marks)	
(a)	What are the key challenges in gathering and managing software requirements? Suggest solutions to overcome these challenges.	CO1
	OR	
(b)	What are the key challenges faced during system testing? Suggest strategies to overcome them and ensure efficient system validation.	CO6